

Report

DevOps Environment Setup

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Introduction

Rahul, Senior DevOps Engineer at TechCorp, required assistance in setting up secure and monitored development environments for two new developers, Sarah and Mike. As a fresher DevOps engineer, I implemented system monitoring, user management, and automated backup procedures to meet operational and security standards.

Task 1: System Monitoring Setup

Objective: Ensure visibility into CPU, memory, processes, and disk usage for performance and capacity planning.

Steps Implemented:

1. Installed monitoring tools

```
sudo apt-get install htop
sudo apt-get install nmon
```

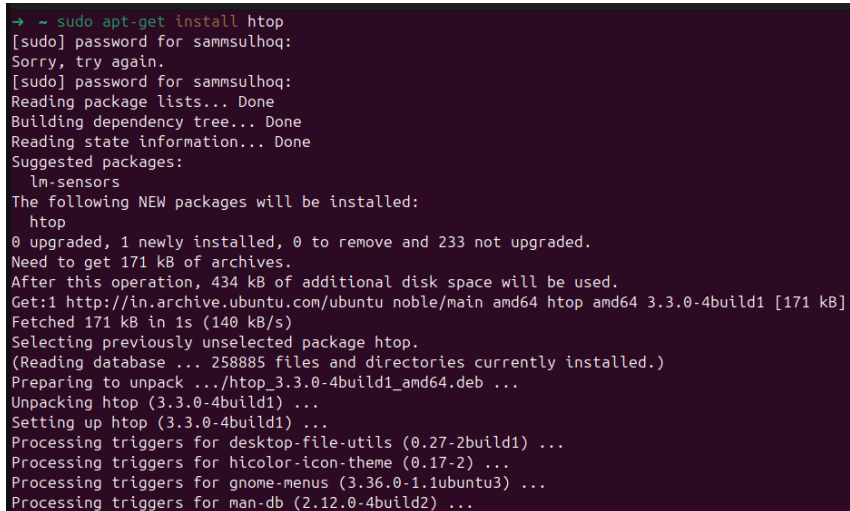
2. Monitored CPU and memory usage with **htop** and **nmon**.
3. Checked disk usage:

```
df -h
du -sh /home/*
```

4. Logged outputs into a file:

```
sudo sh -c 'htop -n 1 > /var/log/system_monitor.log'
sudo sh -c 'df -h >> /var/log/system_monitor.log'
```

Screenshots:



```
→ ~ sudo apt-get install htop
[sudo] password for sammsulhoq:
Sorry, try again.
[sudo] password for sammsulhoq:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  lm-sensors
The following NEW packages will be installed:
  htop
0 upgraded, 1 newly installed, 0 to remove and 233 not upgraded.
Need to get 171 kB of archives.
After this operation, 434 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble/main amd64 htop amd64 3.3.0-4build1 [171 kB]
Fetched 171 kB in 1s (140 kB/s)
Selecting previously unselected package htop.
(Reading database ... 258885 files and directories currently installed.)
Preparing to unpack .../htop_3.3.0-4build1_amd64.deb ...
Unpacking htop (3.3.0-4build1) ...
Setting up htop (3.3.0-4build1) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

```

→ ~ sudo apt-get install nmon
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  nmon
0 upgraded, 1 newly installed, 0 to remove and 233 not upgraded.
Need to get 71.4 kB of archives.
After this operation, 184 kB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 nmon amd64 16p+debian-1 [71.4 kB]
Fetched 71.4 kB in 1s (58.3 kB/s)
Selecting previously unselected package nmon.
(Reading database ... 258895 files and directories currently installed.)
Preparing to unpack .../nmon_16p+debian-1_amd64.deb ...
Unpacking nmon (16p+debian-1) ...
Setting up nmon (16p+debian-1) ...
Processing triggers for man-db (2.12.0-4build2) ...

```

```

→ ~ df -h

```

Filesystem	Size	Used	Avail	Use%	Mounted on
tmpfs	3.9G	2.8M	3.9G	1%	/run
/dev/nvme0n1p2	937G	53G	837G	6%	/
tmpfs	20G	95M	20G	1%	/dev/shm
tmpfs	5.0M	8.0K	5.0M	1%	/run/lock
efivarfs	246K	171K	71K	71%	/sys/firmware/efi/efivars
/dev/nvme0n1p1	1.1G	6.3M	1.1G	1%	/boot/efi
tmpfs	3.9G	128K	3.9G	1%	/run/user/1000

```

→ ~ du -sh /home/*
17G    /home/sammsulhoq

```

```

→ ~ sudo sh -c 'df -h >> /var/log/system_monitor.log'
→ ~ sudo sh -c 'htop -b -n 1 > /var/log/system_monitor.log'
htop: invalid option -- 'b'
→ ~ sudo sh -c 'htop -n 1 > /var/log/system_monitor.log'

```

Task 2: User Management and Access Control

Objective: Create secure accounts and isolated workspaces for Sarah and Mike.

Steps Implemented:

1. Created user accounts:

```
sudo adduser sarah  
sudo adduser mike
```

2. Created dedicated workspaces:

```
mkdir /home/sarah/workspace  
mkdir /home/mike/workspace
```

3. Set ownership and permissions:

```
chown sarah:sarah /home/sarah/workspace  
chmod 700 /home/sarah/workspace  
chown mike:mike /home/mike/workspace  
chmod 700 /home/mike/workspace
```

4. Enforced password policy (expire every 30 days):

```
sudo chage -M 30 sarah  
sudo chage -M 30 mike
```

Screenshots:

```
→ ~ sudo adduser sarah
info: Adding user `sarah' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `sarah' (1001) ...
info: Adding new user `sarah' (1001) with group `sarah (1001)' ...
info: Creating home directory `/home/sarah' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: password updated successfully
Changing the user information for sarah
Enter the new value, or press ENTER for the default
    Full Name []: Sarah Muller
    Room Number []: 101
    Work Phone []: 1234567890
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
info: Adding new user `sarah' to supplemental / extra groups `users' ...
info: Adding user `sarah' to group `users' ...
```

```
→ ~ sudo adduser mike
info: Adding user `mike' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `mike' (1002) ...
info: Adding new user `mike' (1002) with group `mike (1002)' ...
info: Creating home directory `/home/mike' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: password updated successfully
Changing the user information for mike
Enter the new value, or press ENTER for the default
    Full Name []: Mike Cooper
    Room Number []: 102
    Work Phone []: 0987654321
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
info: Adding new user `mike' to supplemental / extra groups `users' ...
info: Adding user `mike' to group `users' ...
```

Name ^

 mike ✖

 sammsulhoq

 sarah ✖

```
→ ~ sudo chown sarah:sarah /home/sarah/workspace
→ ~ sudo chmod 700 /home/sarah/workspace
→ ~ sudo chown mike:mike /home/mike/workspace
→ ~ sudo chmod 700 /home/mike/workspace
chmod: invalid mode: 'chmod'
Try 'chmod --help' for more information.
→ ~ sudo chmod 700 /home/mike/workspace
```

```
→ ~ sudo chage -M 30 sarah
→ ~ sudo chage -M 30 mike
```

Task 3: Backup Configuration for Web Servers

Objective: Automate backups for Apache (Sarah) and Nginx (Mike).

Steps Implemented:

1. Created backup directory:

```
mkdir /backups
```

2. Backup scripts:

Apache (Sarah)

```
tar -czf /backups/apache_backup_$(date +%F).tar.gz /etc/httpd/  
/var/www/html/
```

Nginx (Mike)

```
tar -czf /backups/nginx_backup_$(date +%F).tar.gz /etc/nginx/  
/usr/share/nginx/html/
```

3. Scheduled cron jobs (every Tuesday at 12:00 AM):

```
crontab -e
```

Added lines:

```
0 0 * * 2 tar -czf /backups/apache_backup_$(date +%F).tar.gz /etc/apache2/  
/var/www/html/  
0 0 * * 2 tar -czf /backups/nginx_backup_$(date +%F).tar.gz /etc/nginx/  
/usr/share/nginx/html/
```

4. Verified backups:

```
tar -tzf /backups/apache_backup_YYYY-MM-DD.tar.gz  
tar -tzf /backups/nginx_backup_YYYY-MM-DD.tar.gz
```

```
→ ~ sudo mkdir /backups  
[sudo] password for sammsulhoq:  
□
```



```
tar: Exiting with failure status due to previous errors
→ ~ sudo tar -czf /backups/apache_backup_$(date +%F).tar.gz /etc/apache2/ /var/www/html/
tar: Removing leading '/' from member names
tar: Removing leading '/' from hard link targets
→ ~
```

```
→ ~ sudo tar -czf /backups/nginx_backup_$(date +%F).tar.gz /etc/nginx/ /usr/share/nginx/html/
tar: Removing leading '/' from member names
tar: Removing leading '/' from hard link targets
→ ~
```

```
→ ~ crontab -e
no crontab for sammsulhoq - using an empty one
crontab: installing new crontab
→ ~
```
